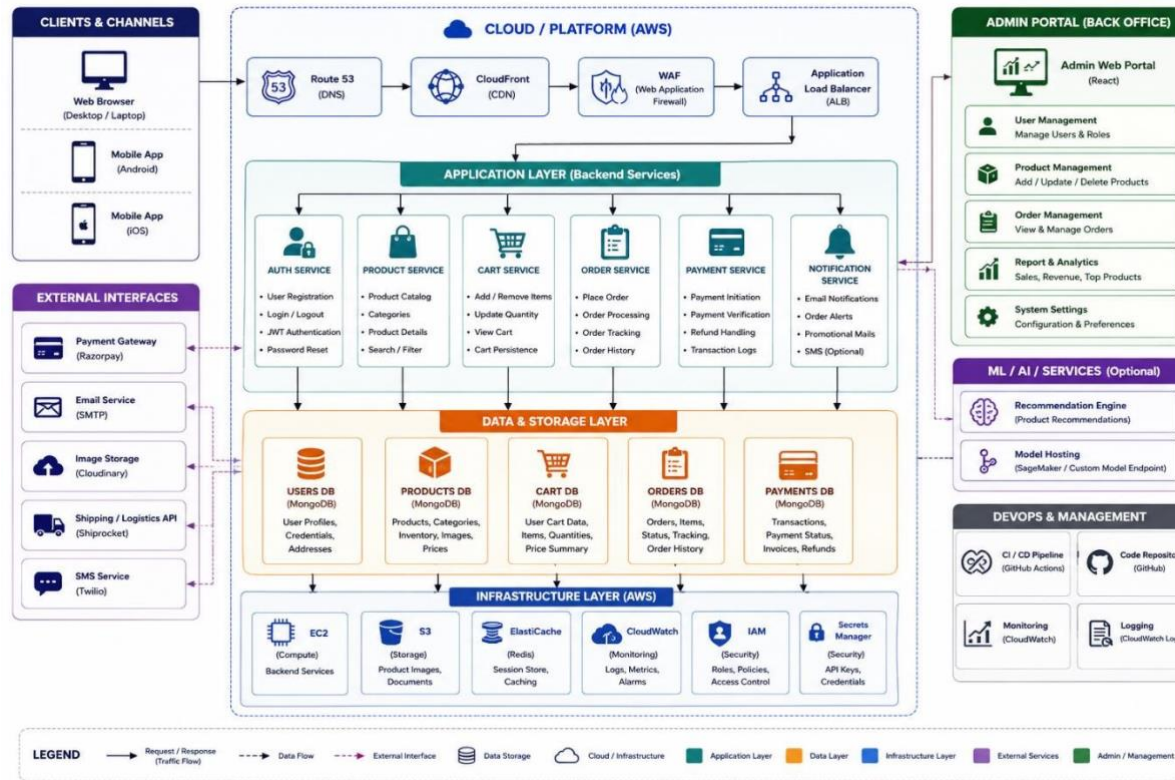


## Project Design Phase-II Technology Stack (Architecture & Stack)

|               |                                |
|---------------|--------------------------------|
| Date          | 18 June 2026                   |
| Prepared By   | Mohit Varma                    |
| Project Name  | ShopEZ: E-Commerce Application |
| Maximum Marks | 4 Marks                        |

### Technical Architecture for ShopEZ:



**Table-1 : Components & Technologies:**

| S.No | Component                       | Description                                               | Technology                            |
|------|---------------------------------|-----------------------------------------------------------|---------------------------------------|
| 1.   | User Interface                  | Enables users to browse products and place orders.        | React.js, Vite, Tailwind CSS          |
| 2.   | Application Logic-1             | Processes user authentication, cart, and order functions. | Node.js, Express.js, JWT, bcryptjs    |
| 3.   | Application Logic-2             | Handles product creation, updates, and retrieval.         | Node.js, Express.js, Mongoose         |
| 4.   | Application Logic-3             | Provides tools for managing products and orders.          | Node.js, Express.js, MVC Architecture |
| 5.   | Database                        | Stores user, product, cart, and order information.        | MongoDB (Mongoose ODM)                |
| 6.   | Cloud Database                  | Maintains application data on the cloud.                  | MongoDB Atlas                         |
| 7.   | File Storage                    | Stores and serves product images.                         | URL-based Image Storage               |
| 8.   | External API-1                  | No external services integrated.                          | N/A                                   |
| 9.   | External API-2                  | No third-party APIs utilized.                             | N/A                                   |
| 10.  | Machine Learning Model          | Not required for this application.                        | N/A                                   |
| 11.  | Infrastructure (Server / Cloud) | Supports deployment and backend operations.               | Node.js Server, MongoDB Atlas         |

**Table-2: Application Characteristics:**

| S.No | Characteristics          | Description                                             | Technology                     |
|------|--------------------------|---------------------------------------------------------|--------------------------------|
| 1.   | Open-Source Frameworks   | Developed using open-source web technologies.           | React.js, Node.js, Express.js  |
| 2.   | Security Implementations | Protects user accounts and application data.            | JWT Authentication, bcrypt     |
| 3.   | Scalable Architecture    | Separates frontend, backend, and database layers.       | React.js, Express.js, MongoDB  |
| 4.   | Availability             | Uses cloud services for reliable access.                | MongoDB Atlas, Node.js         |
| 5.   | Performance              | Ensures smooth communication between system components. | MongoDB, Express.js, REST APIs |